Bug Reports — HW3: UART / CLI

Author: Alexandra Holoborodko · Embedded QA course · Source: UART CLI Homework — Test Report (5.10.2026)

Robustness defects found without inspecting the source code (19 test cases). Boundary analysis: speed −1, 0, 1, 254, 255, 256; timeout 0, 1, 2, 59, 60, 61; input length 0, 1, 39, 40, 41, 256. Behaviour changes at length 41 (SYSTEM FAULT). Build not approved for release.

Environment: Wokwi simulation, UART CLI Homework Device v2.0, Serial Monitor; black-box

Summary

ID	Title	Severity
BUG-01	SYSTEM FAULT after entering CLI input longer than 40 characters	Critical
BUG-02	No error is shown after CLI input of an invalid command	Major
BUG-03	System saves speed value 255 after entering the command "set speed 256"	Major
BUG-04	No error is returned and timeout is modified after entering "set timeout 0"	Major
BUG-05	OK is returned after entering special characters only	Major
BUG-06	Uppercase command STATUS is accepted	Major
BUG-07	SYSTEM FAULT after entering the incomplete command "set" while config is DIRTY	Critical

BUG-01 — SYSTEM FAULT after entering CLI input longer than 40 characters

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter a CLI input longer than 40 characters (helphelphelphelphelphelphelphelphelphelphelp). 4. Check the output.
Expected	ERROR: command too long Long input must not cause a reboot, hang or parser corruption.
Actual	SYSTEM FAULT, the system is rebooting
Reproducibility	Always
Severity	Critical

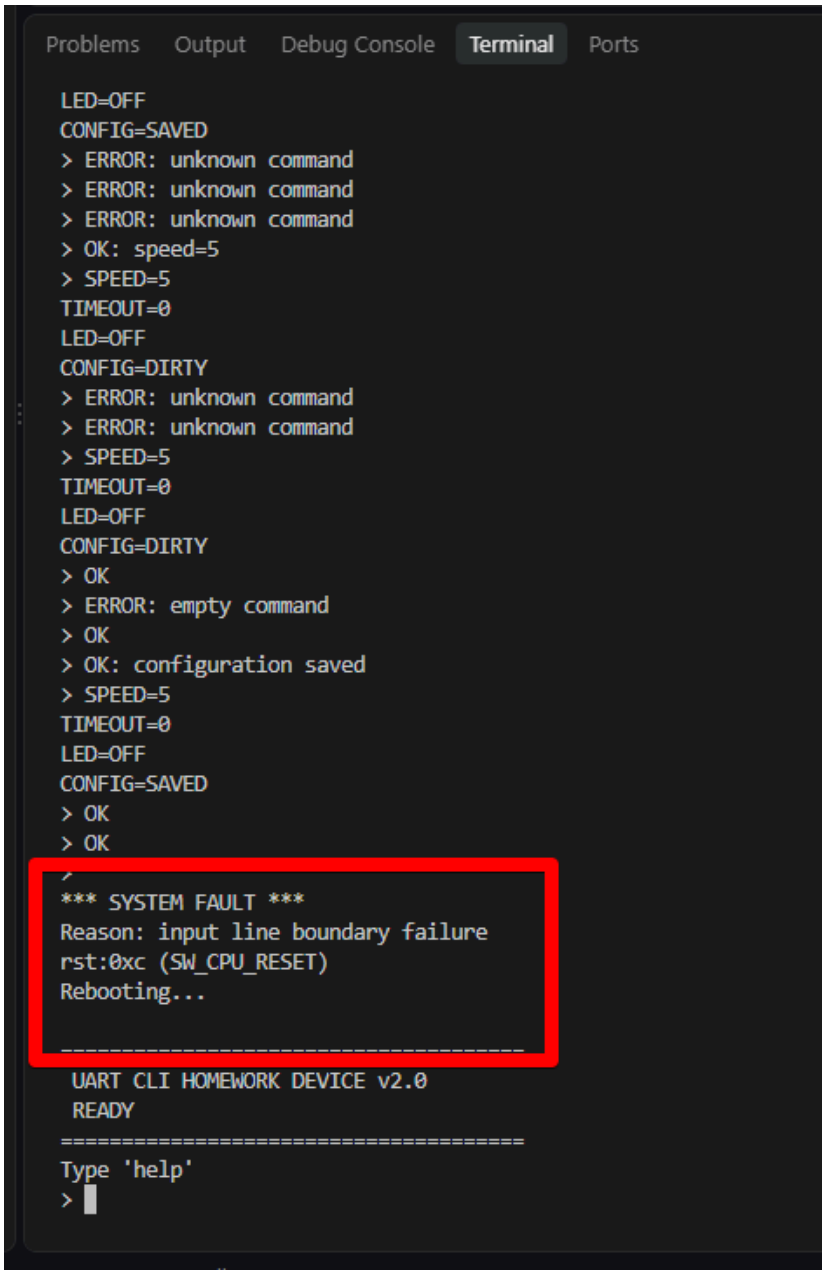


Figure: SYSTEM FAULT on long input

BUG-02 — No error is shown after CLI input of an invalid command

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter the CLI input "set speed5" (without a space) and "тесткоманда" (Cyrillic for "test command"). 4. Check the output.
Expected	ERROR: unknown command
Actual	set speed5 -> OK: speed=5 тесткоманда -> OK
Reproducibility	Always
Severity	Major

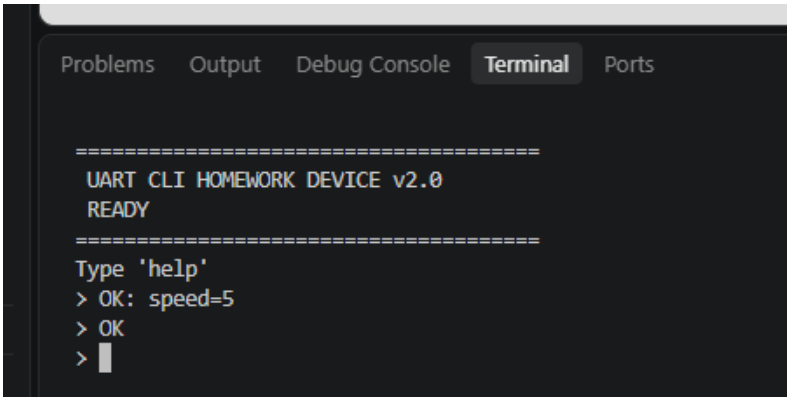


Figure: Invalid commands accepted

BUG-03 — System saves speed value 255 after entering the command "set speed 256"

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter the CLI input "set speed 256". 4. Check the output.
Expected	ERROR is returned and speed is not modified
Actual	OK: speed=255; config=dirty
Reproducibility	Always
Severity	Major

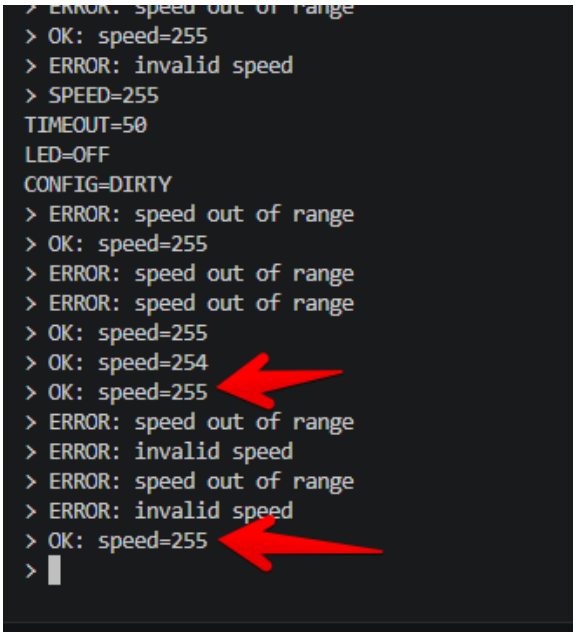


Figure: speed 256 clamped to 255 and accepted

BUG-04 — No error is returned and timeout is modified after entering "set timeout 0"

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter the CLI input "set timeout 0". 4. Check the output.
Expected	ERROR is returned and timeout is not modified
Actual	OK: timeout=0
Reproducibility	Always
Severity	Major

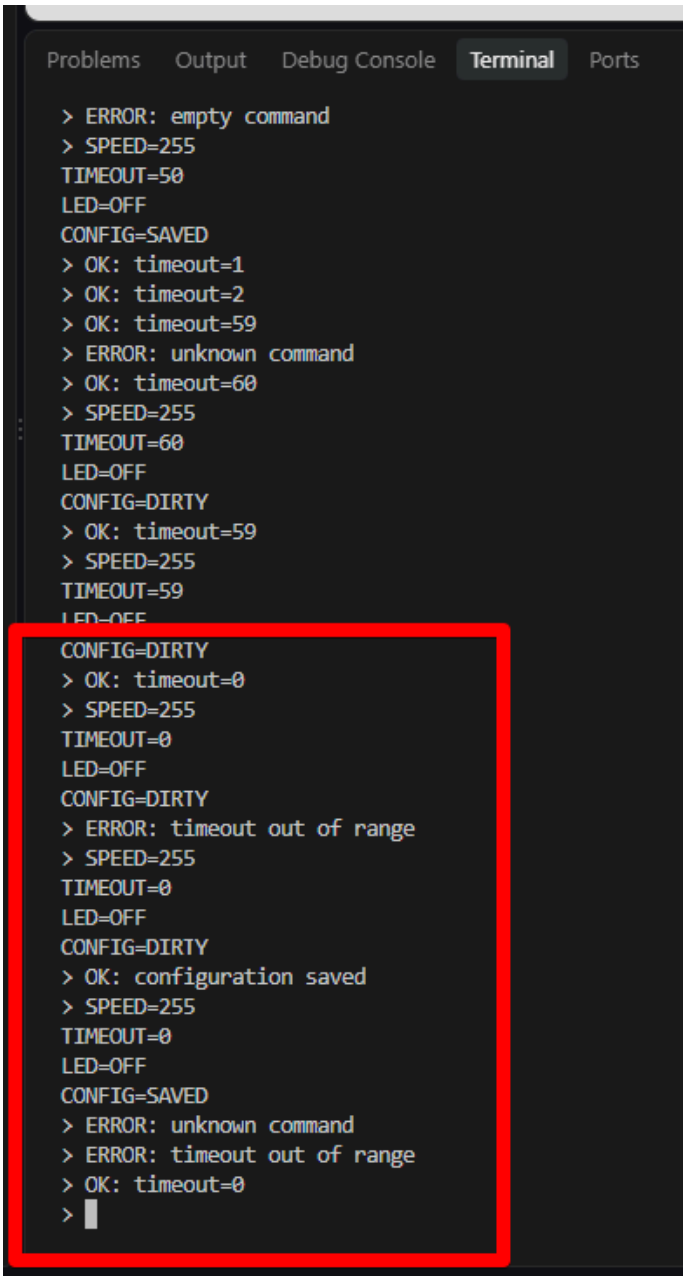


Figure: timeout=0 accepted

BUG-05 — OK is returned after entering special characters only

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter a CLI input made of special characters only: #@*& 4. Check the output.
Expected	ERROR is returned
Actual	> OK
Reproducibility	Always
Severity	Major

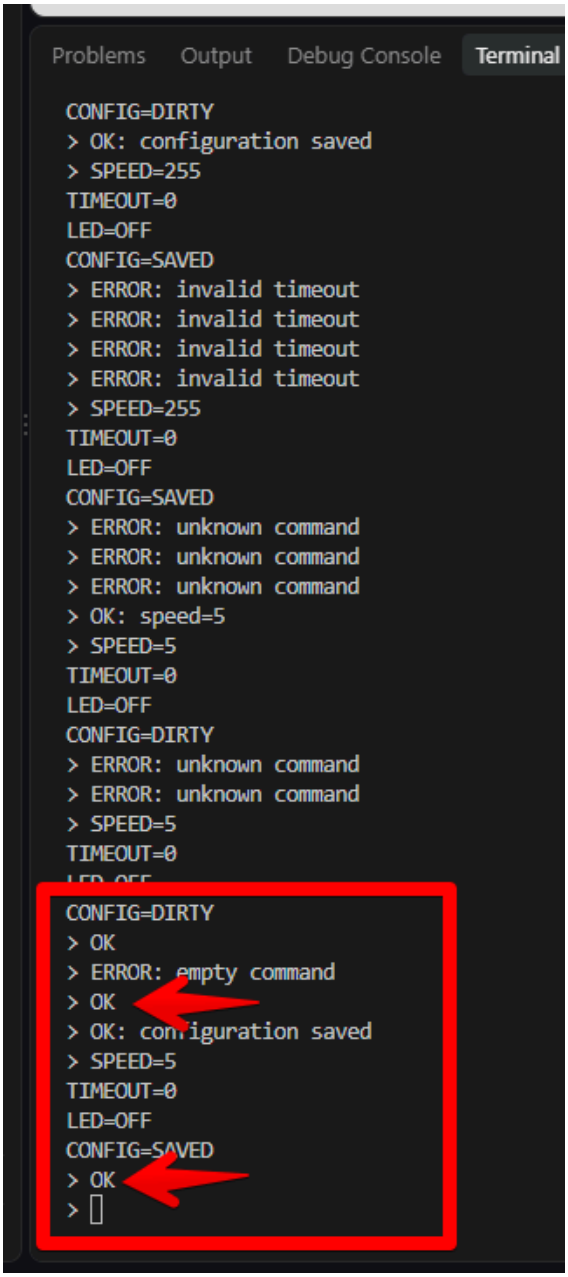


Figure: Special-character input returns OK

BUG-06 — Uppercase command STATUS is accepted

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter the CLI input STATUS (upper case). 4. Check the output.
Expected	ERROR: unknown command (commands are case sensitive)
Actual	STATUS is accepted
Reproducibility	Always
Severity	Major

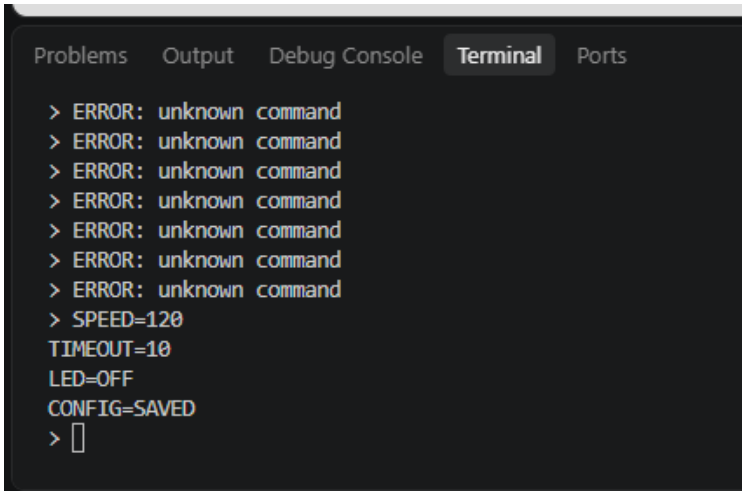


Figure: Terminal output for the STATUS test

BUG-07 — SYSTEM FAULT after entering the incomplete command "set" while config is DIRTY

Steps	1. Turn on the Wokwi simulation. 2. Open the Serial Monitor. 3. Enter the CLI input "set speed 200" or "set timeout 45" (config becomes DIRTY). 4. Enter the CLI input "set". 5. Check the output.
Expected	ERROR is returned
Actual	SYSTEM FAULT, the system is rebooting (Reason: parser state mismatch while configuration is dirty)
Reproducibility	Always
Severity	Critical

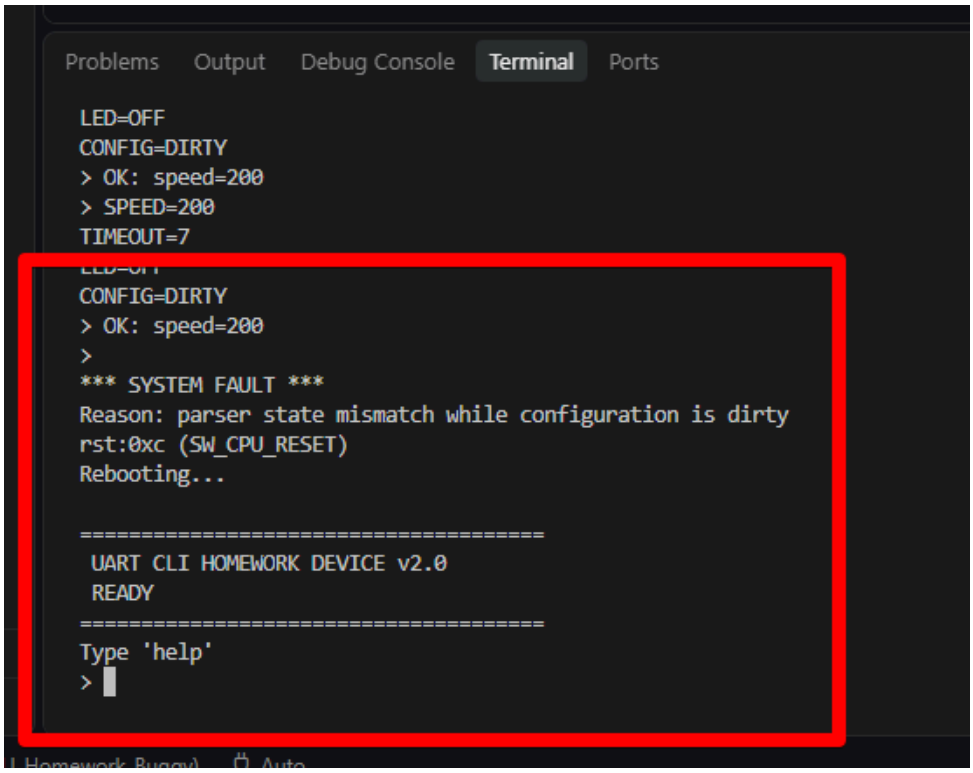


Figure: SYSTEM FAULT in DIRTY configuration